

Section 1**10 marks****Attempt Questions 1 -10****Allow about 15 mins for this section**Use multiple choice answer sheet for Questions 1 -10

1. $\sin 3\alpha \cos 3\alpha =$

- A) $\frac{1}{2} \sin 6\alpha$ B) $9 \sin \alpha \cos \alpha$ C) $\frac{1}{2} \sin 6\alpha$ D) $2 \sin 2\alpha$

2. The point R divides the interval joining P $(-3, 6)$ and Q $(6, -6)$ externally in the ratio 2:1.

Which of these are coordinates of R?

- A) $(3, -2)$ B) $(15, -18)$ C) $(0, 2)$ D) $(-12, 18)$

3. The polynomial $3x^3 - 4x^2 + 2x - 1 = 0$ has roots α , β , and γ .

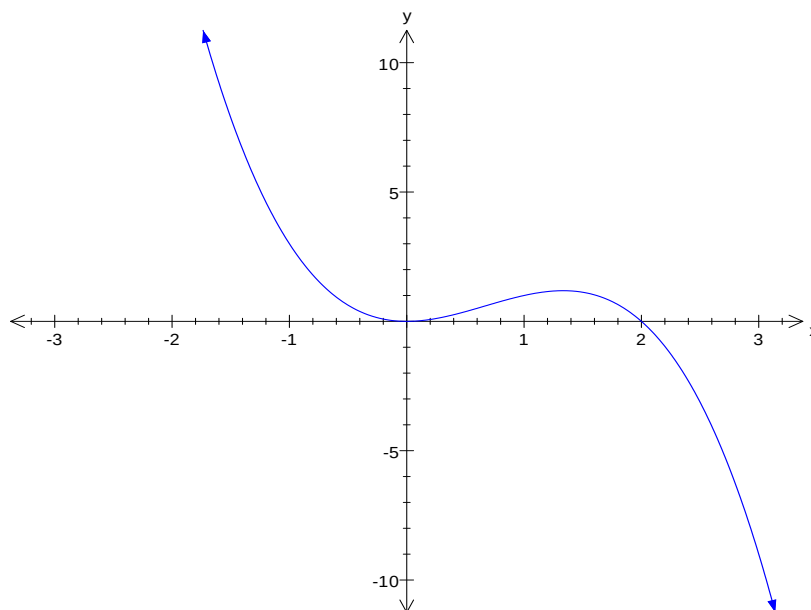
What is the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$?

- A) $-\frac{1}{2}$ B) 2 C) -2 D) -4

4. Which of the following are solutions of $\frac{x}{2x-1} \geq 0$?

- A) $x \geq 0$ B) $0 \leq x < \frac{1}{2}$
C) $x \geq -\frac{1}{2}$ D) $x \leq 0, x > \frac{1}{2}$

5. Which of the following could be the equation of the polynomial shown below?



- A) $P(x) = x^2(2 - x)$ B) $P(x) = x(2 - x)$
 C) $P(x) = (x - 2)x^2$ D) $P(x) = x(x - 2)$
6. What is the derivative of $y = \tan^{-1} \frac{1}{3x}$?

- A) $\frac{9x^2}{1+9x^2}$ B) $\frac{3x}{3+x^2}$ C) $-\frac{3x}{1+9x^2}$ D) $-\frac{3}{9x^2+1}$

7. A particle is moving in simple harmonic motion with displacement x . Its velocity v is given by $v^2 = \frac{1}{4}(16 - x^2)$. What are the amplitude and period of the motion?

- A) amplitude 4, period 4π B) amplitude 16, period $\frac{1}{2\pi}$
 C) amplitude 4, period $\frac{\pi}{4}$ D) amplitude $\frac{1}{2}$, period $\frac{\pi}{4}$

8. $\int \sin^2 x dx =$

A) $\frac{1}{2}x - \frac{1}{4}\cos 2x + c$

B) $\frac{1}{2}x - \frac{1}{4}\sin 2x + c$

C) $\frac{1}{2}x + \frac{1}{4}\cos 2x + c$

D) $\frac{1}{2}x + \frac{1}{4}\sin 2x + c$

9. Evaluate $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{2x}$

A) $\frac{2}{3}$

B) $\frac{1}{6}$

C) $\frac{3}{2}$

D) 6

10. What is the coefficient of x^4 in the expansion of $(1 - x)(1 + x)^9$?

A) 9

B) -84

C) 42

D) 126

Section II

60 Marks

Attempt Questions 11 – 14

Allow 1 hours 45 minutes for this section.

Answer the questions in the own writing booklets provided.

Start each question on a new page.

All necessary working should be shown in every question

Question 11 (15 marks)

Marks

(a) Use the substitution $t = \tan \frac{x}{2}$ to show that $\frac{1+\cos x+\sin x}{1-\cos x+\sin x} = \cot \frac{x}{2}$ **3**

(b) Use the substitution $u = 5 - x$, to find the exact value of $\int_1^5 x\sqrt{5-x}dx$ **3**

(c) Let $f(x) = \frac{1}{\sqrt{x-1}}$

(i) Find the domain and range of $f(x)$. **2**

(ii) Find an expression for the inverse function $f^{-1}(x)$. **2**

(iii) Find the domain and range of the inverse function. **1**

(d) Prove by mathematical induction that,

$$\left(1 - \frac{1}{2^2}\right) \times \left(1 - \frac{1}{3^2}\right) \times \left(1 - \frac{1}{4^2}\right) \times \dots \dots \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n} \text{ for all integer } n \geq 2 \quad \mathbf{4}$$

Question 12 (15 marks) Start this question in a new booklet. **Marks**

- (a) A metal rod is taken from a freezer at -10°C into a room where the air temperature is 25°C .

The rate at which the rod warms follows Newton's law, that is

$$\frac{dT}{dt} = -k(T - 25)$$

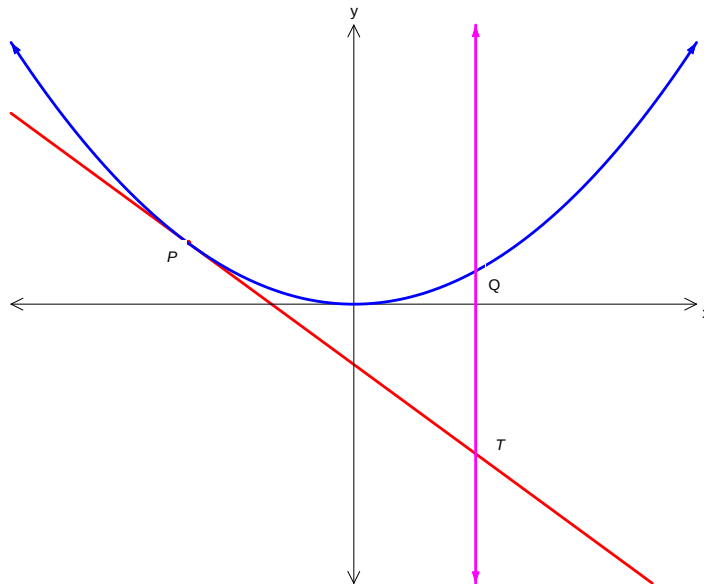
Where k is a positive integer, time t is measured in minutes and temperature T is measured in degrees Celcius.

- (i) Show that $T = 25 - Ae^{-kt}$ is a solution of the differential equation **2**
 $\frac{dT}{dt} = -k(T - 25)$, and find the value of A .
- (ii) If the temperature of the rod reaches 5°C in 80 minutes, find the exact **2**
value of k .
- (iii) Find the temperature of the rod after a further hour. **1**
- (iv) Sketch the graph of the temperature of the rod against time. **1**

Question 12 continues on the next page.

Question 12 continued**Marks**

(b) The points $P(6p, 3p^2)$ and $Q(6q, 3q^2)$ lie on the parabola $x^2 = 12y$.



- (i) Show that the equation of the tangent to the parabola at P is $y = px - 3p^2$ **2**
- (ii) The tangent at P and the line passing through Q parallel to the y -axis intersect at T . Show that the coordinates of T are $(6q, 6pq - 3p^2)$. **2**
- (iii) Find the coordinates of M , the midpoint of PT . **1**
- (iv) Find the Cartesian equation of the locus of M when $pq = -2$. **1**
- (c) (i) Show that the equation $\ln x - \cos x = 0$ has a root between $x = 1$ and $x = 2$. **1**
- (ii) By taking $x = 1.2$ as the first approximation, use one step of Newton's method to find a better approximation to this root. **2**
- Answer to 2 decimal places.

Question 13 (15 marks) Start this question in a new booklet.

Marks

(a) Sketch the function $y = \pi \cos^{-1} \frac{x}{2}$ **2**

(b) The equation of motion for a particle undergoing simple harmonic motion is

$$\ddot{x} = -n^2 x$$

Where x (metres) is the displacement of the particle from the origin at time t (seconds) and n is a positive constant.

(i) Verify that $x = a \sin(nt + \alpha)$, where n , t , and α are constants, is a solution of the equation of motion. **1**

(ii) The particle is initially stationary at $x = 2$ and the function has a period of 3 seconds, find the values of n , a , and α . **3**

(c) Consider the function $f(x) = xe^{x^2}$

(i) Show that the gradient of this function is always positive. **2**

(ii) Find any points of inflexion. **1**

(iii) Sketch the graph of $y = f(x)$ **1**

Question 13 continues on the next page.

Question 13 continued

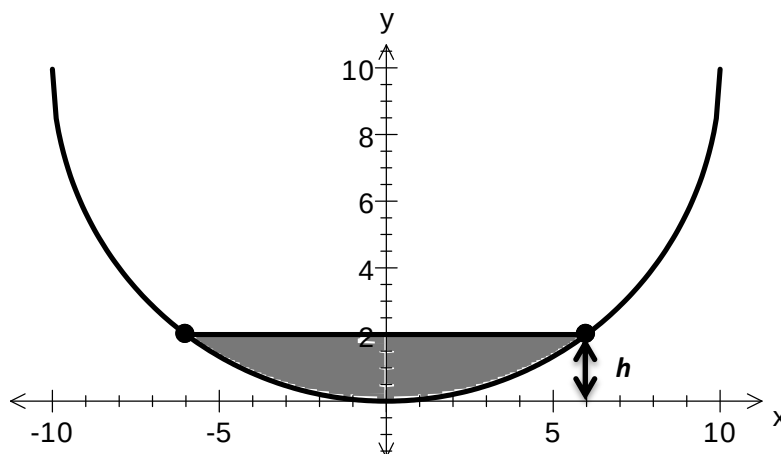
Marks

(d) A hemispherical bowl of radius 10cm is being filled with water at a constant rate of 25 cm^3 per minute.

- (i) By finding the volume of revolution formed by rotating the semi circle **3**

$y = 10 - \sqrt{100 - x^2}$ about the **y-axis** from $y = 0$ to $y = h$, show that the volume (V) of the water in terms of its depth (h) is given by

$$V = \pi \left(\frac{-h^3}{3} + 10h^2 \right).$$



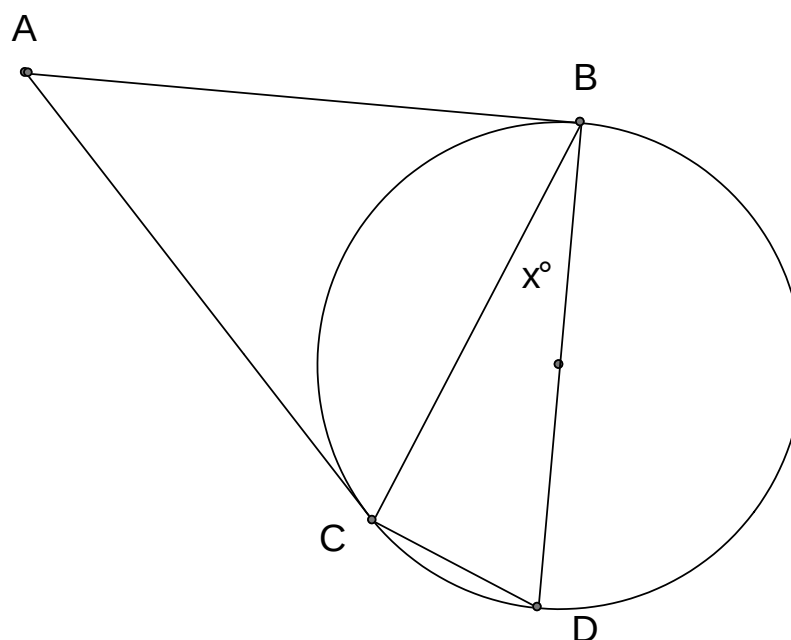
- (ii) At what rate is the water rising when the depth is 5cm. Answer to **2**
2 decimal places.

Question 14 (15 marks) Start this question in a new booklet.

- (a) AB and AC are tangents to a circle. D is a point on the circle such that $\angle BDC = \angle BAC$ and $\angle BAC = 2 \times \angle DBC$. Let $\angle DBC = x$.

(i) Show that DB is a diameter. 4

(ii) Show that $BC = AB$. 1



(b) Consider the expansion:

$$(1+x)^{2n+1} = \binom{2n+1}{0} + \binom{2n+1}{1}x + \binom{2n+1}{2}x^2 + \dots + \binom{2n+1}{2n+1}x^{2n+1}$$

(i) Show that $\sum_{r=1}^{2n+1} \binom{2n+1}{r} = 2 \times 4^n - 1$ 2

(ii) By differentiating, the original expansion above with respect to x , or otherwise, show that

$$\binom{2n+1}{1} - 2\binom{2n+1}{2} + 3\binom{2n+1}{3} - \dots + (2n+1)\binom{2n+1}{2n+1} = 0$$
 3

Question 14 continues on the next page.

Question 14 continued**Marks**

- (c) A cannonball is fired from a cannon on top of an 80 metre high cliff down on to a ship in the sea below with velocity 40 metres per second at an angle of inclination θ to the horizontal. The equations of motion of the cannonball are:

$$x = 35t\cos\theta \text{ and } y = -4.9t^2 + 35t\sin\theta + 80 \text{ (Do NOT prove this).}$$

- (i) By eliminating t show that the Cartesian equation of the path of the cannonball is given by $y = \frac{-\sec^2\theta}{250}x^2 + x\tan\theta + 80$ **2**
- (ii) In order to hit the ship, the cannonball must land 150 metres from the base of the cliff. For what values of θ will this occur? **3**

End of Paper

MC

Q1 $\sin 2A = 2 \sin A \cos A$

let $A = 3x$ $\sin 6x = 2 \sin 3x \cos 3x$

$\sin 3x \cos 3x = \frac{1}{2} \sin 6x$

(A)

Q2. Use ratio $2:-1$

$R = \left(\frac{2 \times 6 - 1 \times 3}{2 - 1}, \frac{2 \times -6 - 1 \times 6}{2 - 1} \right)$

$= (15, -18)$

(B)

Q3. $\alpha + \beta + \gamma = \frac{4}{3}$

$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{2}{3}$

$\alpha\beta\gamma = \frac{1}{3}$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma}$

$= \frac{\frac{2}{3}}{\frac{1}{3}}$

$= 2$

(B)

Q4. $\therefore x(2x-1) \geq 0$

$(x-0)(x-\frac{1}{2}) \leq 0$

$(2x-1)(x-\frac{1}{2}) \leq 0$

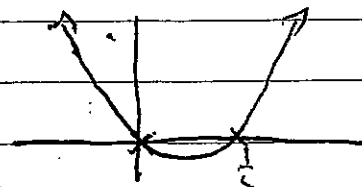
$(x-\frac{1}{2})(x-\frac{1}{2}) \leq 0$

$\frac{1}{2} \leq x \leq \frac{1}{2}$

$x(2x-1) \geq 0$ $x \neq \frac{1}{2}$

$x \leq 0, x > \frac{1}{2}$

(D)



Q5. (A)

Q6. $y = \tan^{-1} \frac{1}{3x}$

$$\frac{dy}{dx} = \frac{1}{1 + \left(\frac{1}{3x}\right)^2} \times -\frac{1}{3} x^{-2}$$

$$= \frac{1}{1 + \frac{1}{9x^2}} \times -\frac{1}{3x^2}$$

$$= -\frac{1}{\frac{3x^2 + \frac{1}{3}}{3}} = -\frac{3}{9x^2 + 1} \quad \text{(D)}$$

Q7. $v^2 = n^2(a^2 - x^2)$ $a = 4$ $n = \frac{1}{2}$

amplitude = 4

$$\text{period} = \frac{2\pi}{n}$$

$$= 4\pi$$

(A)

Q8. (B)

Q9. $\frac{1}{52} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{3}}{\frac{x}{3}} = \frac{1}{6} \times 1$

$$= \frac{1}{6} \quad \text{(B)}$$

Q10. $(1-x)({}^9C_0 + {}^9C_1x + {}^9C_2x^2 + {}^9C_3x^3 + {}^9C_4x^4 + \dots)$

$$\text{Coeff of } x^4 = 1 \times {}^9C_4 - 1 \times {}^9C_3$$

$$= 126 - 84$$

(C)

$$Q11(a) t = \tan \frac{x}{2} \quad \sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$LHS = 1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}$$

$$= \frac{1 - \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}}$$

$$= \frac{\cancel{1+t^2} + \cancel{1-t^2} + 2t}{\cancel{1+t^2}}$$

$$= \frac{1+t^2 - (1-t^2) + 2t}{(1+t^2)}$$

$$= \frac{2+2t}{2t+2t^2}$$

$$= \frac{2(1+t)}{2t(1+t)}$$

$$= \frac{1}{t}$$

$$= \cot \frac{x}{2}$$

$$(b) \int_1^5 x \sqrt{5-x} \, dx$$

$$= \int_4^0 (5-u) u^{1/2} (-du)$$

$$= \int_0^4 5u^{1/2} - u^{3/2} \, du$$

$$= \left[\frac{10}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right]_0^4$$

$$= \left(\frac{10}{3} \times 8 - \frac{2}{5} \times 32 \right) - 0$$

$$= 13 \frac{13}{15} \left(\frac{208}{15} \right)$$

$$u = 5-x \quad x=1 \quad u=4$$

$$x = 5-u \quad x=5 \quad u=0$$

$$\frac{dx}{du} = -1$$

$$dx = -du$$

(b)

(i) $y = \frac{1}{12} x^2$

$$\frac{dy}{dx} = \frac{1}{6} x$$

At $x = 6p$ $\frac{dy}{dx} = p$ ✓

∴ Equation of tangent is

$$y - 3p^2 = p(x - 6p) \quad \checkmark$$

$$y - 3p^2 = px - 6p^2$$

$$y = px - 3p^2 \quad (1)$$

(ii) Line through (1) parallel to y-axis is $x = 6q$. (2) ✓

Point T is where (1) & (2) meet

$$\text{i.e. } y = p \times 6q - 3p^2$$

$$y = 6pq - 3p^2$$

∴ intersection is at T $(6q, 6pq - 3p^2)$

(iii) $M = \left(\frac{6p+6q}{2}, \frac{3p^2+6pq-3p^2}{2} \right)$

$$= (3p+3q, 3pq) \quad \checkmark$$

(iv) When $pq = -2$, y-value of M is -6 is constant
∴ equation of locus of M is $y = -6$. ✓

(c) Let $f(x) = \ln x - \cos x$

$$f(1) = -0.5403 \dots$$

$$f(2) = 1.10929 \dots \checkmark$$

Since $f(1) < 0$, $f(2) > 0$ and $f(x)$ is continuous in this interval, there must be a root.

(ii) $f'(x) = \frac{1}{x} + \sin x$. ✓

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f'(x_0)$$

$$x_1 = 1.2 - \frac{\ln 1.2 - \cos 1.2}{\frac{1}{1.2} + \sin 1.2}$$

$$\frac{1}{1.2} + \sin 1.2$$

$$= 1.30 \quad \checkmark$$

$$(c) f(x) = \frac{1}{\sqrt{x-1}}$$

$$(i) \text{ domain } x > 1 \quad \checkmark$$

$$\text{range } y > 0 \quad \checkmark$$

$$(ii) \text{ let } y = \frac{1}{\sqrt{x-1}}$$

swap x, y for inverse

$$x = \frac{1}{\sqrt{y-1}} \quad \checkmark$$

$$\sqrt{y-1} = \frac{1}{x}$$

$$y-1 = \frac{1}{x^2} \quad x > 0$$

$$y = \frac{1}{x^2} + 1; x > 0 \quad \checkmark \quad \text{must have domain}$$

$$(iii) \text{ ~~domain } x > 0~~ \quad \checkmark$$

$$\text{range } y > 1 \quad \checkmark$$

(d) show true for $n=2$

$$\text{LHS} = 1 - \frac{1}{2^2} = \frac{3}{4}$$

$$\text{RHS} = \frac{2+1}{4} = \frac{3}{4} \quad \checkmark$$

$$\text{LHS} = \text{RHS}$$

if true for $n=k$

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right) = \frac{k+1}{2k} \quad *$$

show true for $n=k+1$

$$\text{a. } \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+2}{2k+2} \quad \checkmark$$

$$\text{LHS} = \left(\frac{k+1}{2k}\right) \left(1 - \frac{1}{(k+1)^2}\right) \quad \text{from } * \quad \checkmark$$

$$= \left(\frac{k+1}{2k}\right) \left(\frac{(k+1)^2 - 1}{(k+1)^2}\right)$$

$$= \left(\frac{k+1}{2k}\right) \left(\frac{k^2 + 2k}{(k+1)^2}\right)$$

$$= \frac{(k+1)}{2k} \times \frac{k(k+2)}{(k+1)^2} \quad \checkmark$$

$$= \frac{k+2}{2k+2} = \text{RHS}$$

correct
 $\Rightarrow$ if no statement
 true for all $n \geq 2$ by principle of
 mathematical induction.

Q 12a) (i) $T = 20 + Ae^{-kt}$

$$\frac{dT}{dt} = kAe^{-kt} \quad \text{but } Ae^{-kt} = (T - 20)$$

$$= -k(T - 20) \quad \checkmark$$

when $t = 0$ $T = -10$.

$$-10 = 20 + Ae^0$$

$$A = -30 \quad \checkmark$$

$$\therefore T = 20 - 30e^{-kt}$$

(ii) when $t = 80$ $T = 5$.

$$5 = 20 - 30e^{-80k} \quad \checkmark$$

$$30e^{-80k} = 15$$

$$e^{-80k} = \frac{1}{2}$$

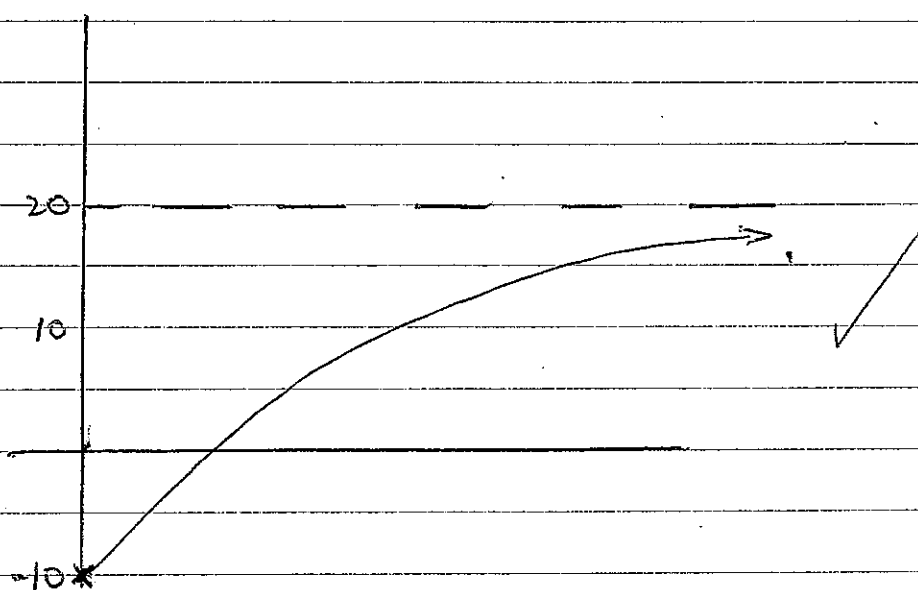
$$-80k = \ln \frac{1}{2}$$

$$k = \frac{\ln \frac{1}{2}}{-80} \quad \checkmark$$

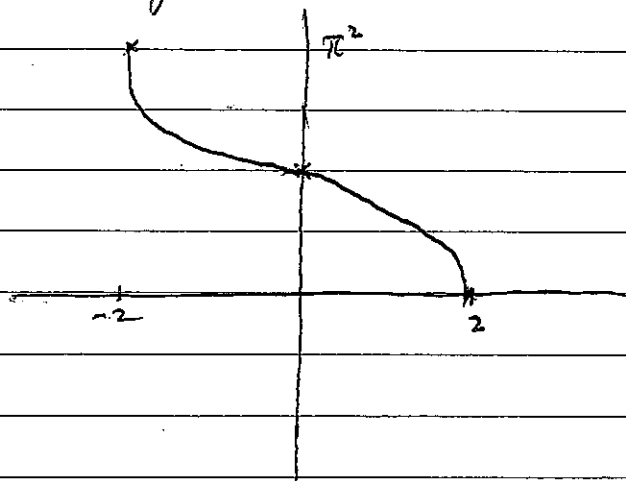
(iii) when $t = 80 + 60$
 $= 140$.

$$T = 20 - 30e^{-k \times 140} \quad \checkmark$$

$$T = 11$$



Q13(a) domain : $-1 \leq \frac{x}{2} \leq 1$ $-2 \leq x \leq 2$
 range : $0 \leq \frac{y}{\pi} \leq \pi$ $0 \leq y \leq \pi^2$



(b)(i) $x = a \sin(nt + \alpha)$
 $\dot{x} = na \cos(nt + \alpha)$
 $\ddot{x} = -n^2 a \sin(nt + \alpha)$
 $= -n^2 x$

(ii) $t=0, \dot{x}=0, x=2$

Initially stationary at $x=2$ means $a=2$

and $2 = 2 \sin(0 + \alpha)$

$\sin \alpha = 1$

$\alpha = \pi/2$

period = 3 seconds

$\frac{2\pi}{n} = 3$

$n = \frac{2\pi}{3}$

(c) $f'(x) = x \cdot 2x e^{x^2} + e^{x^2}$
 $= e^{x^2} (2x^2 + 1)$

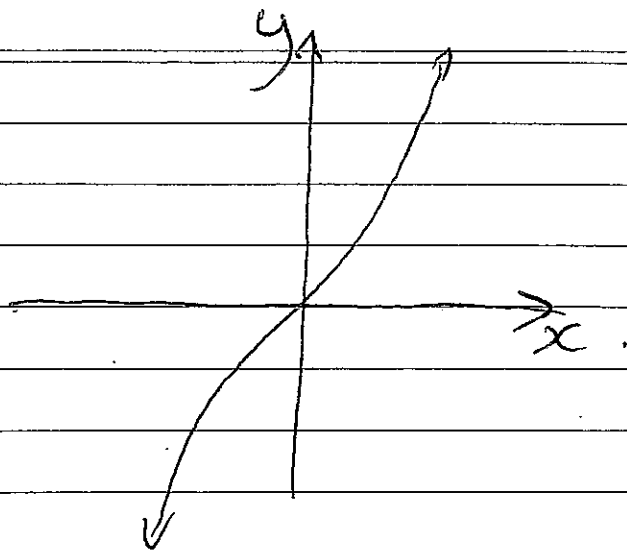
since $e^{x^2} > 0$ and $2x^2 > 0$ $f'(x) > 0$ for all x

$f''(x) = 2x^2 \times [2x e^{x^2}] + e^{x^2} \times 4x + 2x^2 e^{x^2}$
 $= 4x^3 e^{x^2} + 4x e^{x^2} + 2x^2 e^{x^2}$
 $= 2x e^{x^2} (2x^2 + 3)$

$f''(x) = 0$ when $x=0$ (note $e^{x^2} > 0, 2x^2 + 3 > 0$)

At $x=0$ $f'(0) = 1 > 0$

$\therefore$ inflexion at $(0, 0)$



$$d)(i) y = 10 - \sqrt{100 - x^2}$$

$$\sqrt{100 - x^2} = 10 - y$$

$$100 - x^2 = (10 - y)^2$$

$$x^2 = 100 - (10 - y)^2$$

$$x^2 = 100 - (100 - 20y + y^2)$$

$$x^2 = 20y - y^2$$

$$Vol = \pi \int_0^h x^2 dy$$

$$= \pi \int_0^h (20y - y^2) dy$$

$$= \pi \left[10y^2 - \frac{y^3}{3} \right]_0^h$$

$$V = \pi \left(10h^2 - \frac{h^3}{3} \right)$$

$$(ii) \frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$\frac{dV}{dh} = \pi (20h - h^2)$$

$$25 = \pi (20h - h^2) \times \frac{dh}{dt}$$

$$\text{at } h = 5$$

$$25 = \pi (100 - 25) \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{1}{3\pi} \text{ cm/sec}$$

Q4(a)

(i) If $\hat{DBC} = x$, then $\hat{BAC} = 2x$ (given)

$AB = AC$ (tangents from external point are equal)

$\therefore \triangle ABC$ is isosceles

and $\hat{ABC} = \hat{ACB}$ (base $<$ isos $\triangle$)

$$\text{Sum } \therefore \hat{ABC} = \frac{180 - 2x}{2} \quad (\angle \text{ sum } \triangle ABC)$$

$$= 90 - x$$

$$\therefore \hat{ABD} = 90 - x + x$$

$$= 90^\circ$$

$\therefore BD$ is perpendicular to tangent AB and must pass through centre of circle
 $\therefore BD$ is a diameter

(ii) If $\hat{BDC} = \hat{BAC}$

and $\hat{BDC} = 90 - x$ (alternate segment theorem)

$$\text{then } 2x = 90 - x$$

$$3x = 90$$

$$x = 30$$

$$\therefore \hat{CAB} = \hat{ABC} = \hat{BCA} = 60^\circ$$

$\therefore \triangle ABC$ is equilateral

and $BC = AB$

(b) i) sub $x = 1$

$$2^{2n+1} = \binom{2n+1}{0} + \binom{2n+1}{1} + \binom{2n+1}{2} + \dots + \binom{2n+1}{2n+1}$$

$$2 \times 4^n = 1 + \binom{2n+1}{1} + \binom{2n+1}{2} + \dots + \binom{2n+1}{2n+1}$$

$$2 \times 4^n - 1 = \binom{2n+1}{1} + \binom{2n+1}{2} + \dots + \binom{2n+1}{2n+1}$$

$$= \sum_{r=1}^{2n+1} \binom{2n+1}{r}$$

by differentiating both sides

$$(2n+1)(1+x)^{2n} = \binom{2n+1}{1} + 2 \binom{2n+1}{2}x + 3 \binom{2n+1}{3}x^2 + \dots + (2n+1) \binom{2n+1}{2n+1}x^{2n}$$

sub $x = -1$

$$0 = \binom{2n+1}{1} - 2 \binom{2n+1}{2} + 3 \binom{2n+1}{3} - \dots + (2n+1) \binom{2n+1}{2n+1}$$

$$(c) \quad x = 35t \cos \theta \quad (1) \quad y = -4.9t^2 + 35t \sin \theta + 80 \quad (2)$$

$$\text{From (1)} \quad t = \frac{x}{35 \cos \theta}$$

$$\text{sub in (2)} \quad y = -4.9 \left(\frac{x}{35 \cos \theta} \right)^2 + 35 \left(\frac{x}{35 \cos \theta} \right) \sin \theta + 80.$$

$$y = -\frac{4.9}{1225} x^2 \sec^2 \theta + x \tan \theta + 80$$

$$y = -\frac{\sec^2 \theta}{250} x^2 + x \tan \theta + 80.$$

$$(ii) \quad \text{when } y = 0 \quad x = 150.$$

$$0 = -90 \sec^2 \theta + 150 \tan \theta + 80.$$

$$0 = -90 (\tan^2 \theta + 1) + 150 \tan \theta + 80.$$

$$0 = -90 \tan^2 \theta - 90 + 150 \tan \theta + 80$$

$$0 = -90 \tan^2 \theta + 150 \tan \theta - 10.$$

$$9 \tan^2 \theta - 15 \tan \theta + 1 = 0.$$

$$\tan \theta = \frac{15 \pm \sqrt{15^2 - 4 \times 9 \times 1}}{18}.$$

$$\tan \theta = 1.597 \dots \text{ or } 0.0695 \dots$$

$$\theta = 58^\circ \quad \text{or} \quad 4^\circ$$